# Vectors: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

**Solution index**

1 2 3 4 5 6 7 8 9 10

**Solution 1 — Components.** Return →

$A_x = 10\cos 30^\circ = 8.66$ and $A_y = 10\sin 30^\circ = 5.00$. **Check:** $8.66^2 + 5^2 = 100$, so the magnitude is 10.

**Solution 2 — Vector sum.** Return →

The sum is $(2, 6)$. Its magnitude is $\sqrt{4+36} = 6.32$, and its direction is $\tan^{-1}(6/2) = 71.6^\circ$ above the $x$-axis. **Check:** the sum points mostly upward because both vectors have positive $y$ components.

**Solution 3 — Dot product.** Return →

$\mathbf{A}\cdot\mathbf{B} = 2(4) + 3(-1) = 5$. The dot product is positive, so the angle is acute. **Check:** a zero dot product would mean perpendicular vectors.

**Solution 4 — Angle between vectors.** Return →

$\cos\theta = (1)/(1\times 2) = 0.5$, so $\theta = 60^\circ$. **Check:** $\mathbf{B}$ has slope $\sqrt{3}$, which corresponds to 60°.

**Solution 5 — Cross product.** Return →

The $z$-component is $A_x B_y - A_y B_x = 3(4) - 0 = 12$. It is positive, so the vector points out of the page by the right-hand rule. **Check:** rotating from $+x$ to $+y$ is counterclockwise.

**Solution 6 — Relative velocity.** Return →

$\mathbf{v}_{A/B} = (20 - 5, 10 - 25) = (15, -15)$ km/s. The speed is $\sqrt{15^2 + 15^2} = 21.2$ km/s. **Check:** relative velocity is a difference, not a sum.

**Solution 7 — Flux through a surface.** Return →

Flux is $EA\cos\theta = 5 \times 2 \times \cos 60^\circ = 5$. **Check:** if the surface were edge-on to the field, the flux would be zero.

**Solution 8 — Torque.** Return →

The torque is $r_x F_y - r_y F_x = 2(10) - 0 = 20$ N m in the $+z$ direction. **Check:** the force tends to rotate the position vector counterclockwise.

**Solution 9 — Projection.** Return →

The projection is $[(\mathbf{A}\cdot\mathbf{B})/|\mathbf{B}|^2]\mathbf{B} = (6/2)(1,1) = (3,3)$. Its length is $3\sqrt{2} = 4.24$. **Check:** the projection lies along $\mathbf{B}$, as required.

**Solution 10 — Angular momentum direction.** Return →

$\mathbf{r} \times \mathbf{v}$ points in the $+z$ direction, out of the orbital plane. For an isolated Keplerian orbit, angular momentum is conserved, so its direction and magnitude remain constant. **Check:** the orbit is planar because the angular-momentum vector fixes the plane.